

My Magical LED Dice Adventure!

Become a Junior Hardware Engineer & Build Your Own Glowing Dice

1. The Mystery of the Missing Dice

Imagine you are sitting happily on the big floor mat on a rainy Sunday evening. You are playing a super fun game of Ludo with all your cousins, your big brother, and your sweet grandparents. The bright board is laid out right in the middle of everyone. You picked the blue token, and you only need three more steps to reach home and win the whole game!

You shake your hands happily, ready to roll the dice, but oh no! The tiny white dice slips right out of your fingers, tumbles off the sheet, and rolls deep under the heavy wooden storage diwan couch! It is too dark down there to see anything, and nobody can crawl underneath to get it.

Does our game have to stop forever? Are we stuck? No way! We are smart little engineers. Today, we are going to build our very own magical, glowing, electronic LED Dice from scratch!

2. What is a "Surprise Number"?

Have you ever played the paper chit game at school with your friends? You take a clean piece of paper, tear it into small square pieces, and write a secret number or a friend's name on each one. Then, you roll and fold every little paper slip into a tiny, tight ball so they all look exactly the same!

You put all the paper balls into a shiny bowl, mix them around and around, close your eyes tight, and pull one out. Until your fingers open up that paper ball, the number inside is a total secret! That fun surprise is exactly how your new electronic dice picks a "random" number for your game. A computer chip inside switches through numbers lightning-fast and stops on a surprise choice just for you!

3. Meet Your Circuit Friends!

Instead of using tough grown-up words, let's meet the friendly little characters who work together as a team to make your electronic dice light up!



The Energy Tank

(The Coin-Cell Battery)

This is a small, round, silver battery. Think of it like a mini water tank holding all the magical electrical power inside, ready to rush down the wires!



The Gatekeeper

(The Push-Button Switch)

This is a small button that acts like a main door. When you press down, the door swings wide open, letting the power flow through easily!



The Beautiful Light

(The LED Light Matrix)

This is the final stop! When power rushes into this light grid, it instantly turns into a bright, glowing pattern of dots to show your surprise number!

4. Action and Reaction: How It Works

Electronic gadgets work by linking a human action (the **Input**) to a machine reaction (the **Output**). Think of a doorbell: when you press the button (Input), the house makes a loud "Ding-Dong!" sound (Output). Let's see how your dice obeys your commands:

What You Do (Input Action)	What Your Dice Does (Output Result)
You press down on the push button.	The path links together and sends power rushing into the brain chip.
The inner brain chip picks a surprise slip.	Special LED lights instantly turn on to show a happy cluster of dots!
You lift your finger away from the button.	The path opens up, the power stops, and the lights go to sleep safely.

5. The Magic Circle (The Circuit Loop)

Electricity cannot jump across empty spaces in the air. It behaves exactly like a little flowing stream of water that needs a perfect, completely connected loop to move forward. Think of it like walking a **Pradakshina** path around a temple with your family—you must start at the front, walk all the way around, and return to the start without any gaps!

When you build your dice, the power will move out from the positive (+) side of your battery, travel through the button switch, enter the light frame, and go right back down into the negative (-) side to complete the magic circle!

6. Step-by-Step Build Guide

Your building kit uses safe, pre-wrapped magnetic clips that snap together easily. No sharp tools, scissors, or hot tools allowed!

Step 1: Set Up the Power Highway

- ☐ **Find the Plus Mark:** Look closely at the flat, smooth side of your silver battery and find the tiny plus symbol (+).
- ☐ **Get the Red Wire:** Pick up the bright red power wire clip from your box.
- ☐ **Slide and Snap:** Slide the shiny metal mouth of the clip onto the edge of the silver battery until you feel a fun, solid "Click!"

 **What just happened?** You opened up the main gate of your electrical water tank! The power is now safe and flowing down your red wire highway!

Step 2: Connect the Gatekeeper Switch

- ☐ **Find the Button:** Look for the tiny square block that has a bright, colorful round button on top.
- ☐ **Hold the Red Line:** Take the loose metal tip at the end of your red wire.
- ☐ **Push into the Port:** Gently push that metal tip straight into the opening on the left side of your button block.

Step 3: Close the Magic Circle & Test

- ☐ **Connect Blue Wire:** Take the blue wire coming out from the right side of your button.
- ☐ **Plug Into Screen:** Plug that blue wire directly into the base of your central LED block.
- ☐ **Return the Path:** Take the final black return wire coming from the bottom of the LED block, and clip it securely to the flat negative (-) back of your battery.
- ☐ **Fit into Shell:** Press the whole connected assembly gently down into the pre-made slots inside your colorful plastic housing shell.

 **The Grand Magic Test:** Press down firmly on the button with your thumb. Wow! Look at that magic glow! The bright light dots are cycling rapidly and showing a surprise random number every single time you press down!

7. My Junior Engineer Safety Rules

- ☐ **Rule 1: No Sharp or Hot Tools!** Only use the safe click-fit magnetic leads included in your kit.
- ☐ **Rule 2: No Wall Outlets!** Never play with wall plug points. Our safe 3V battery has absolutely zero risk of shock!
- ☐ **Rule 3: Keep it Fun & Local!** Use your dice for games you love like Ludo, Snakes & Ladders, or Paramapadham.
- ☐ **Rule 4: Use Happy Words!** If your light doesn't turn on immediately, don't worry! It's just a cute little sleeping firefly (Jugnu). Let's check our round walking path together to wake it up!

OFFICIAL CERTIFICATE

Junior Hardware Engineer

This certifies that an awesome little creator has successfully completed the **LED Dice Adventure**, built an unbroken magic circuit loop, and mastered the science of inputs, outputs, and surprise numbers!

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